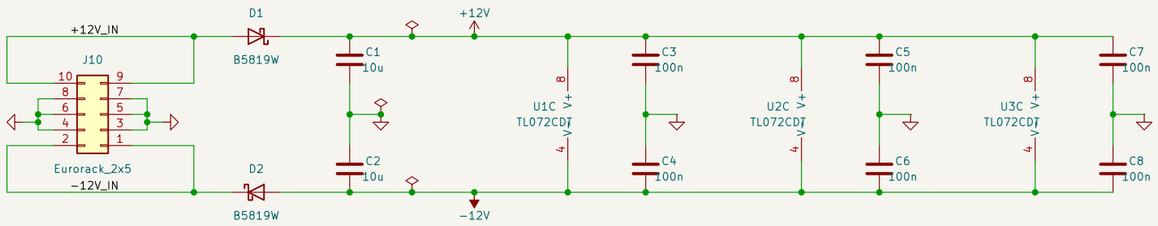
BIG MODULAR LINE IN — S2 schematic: stereo line input to Eurorack level (Bread Modular line_in on +/−12 V). Circuit intent: docs/research.md

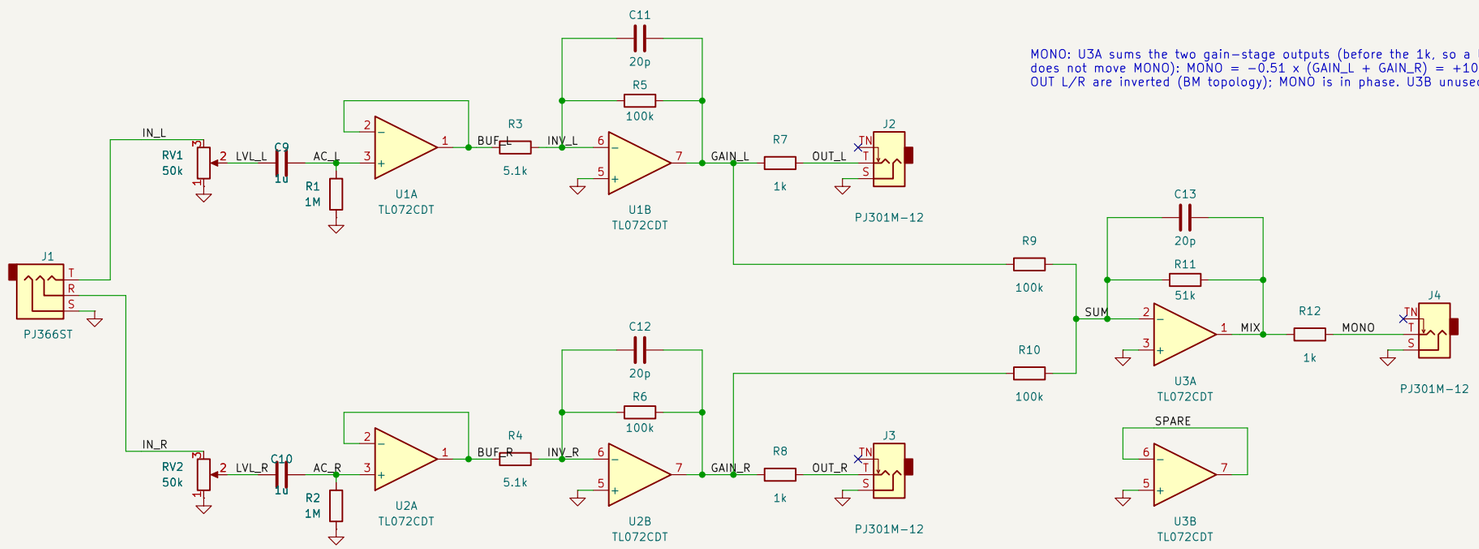
POWER: Eurorack +12V / −12V -> B5819W per rail -> +12V / −12V (about +/−11.6 V). J10 +5V unused.
C3..C8: 100n at every op-amp supply pin; C1/C2: 10u bulk.



PWR_FLAG: #FLG01 marks the Eurorack GND; #FLG02/#FLG03 mark the +12V/−12V rails,
fed from the header only through D1/D2, which ERC cannot see through.

CHANNELS L/R (Bread Modular line_in, ground-referenced): J1 tip/ring -> 50k LEVEL pot -> 1u -> 1M bias
-> follower -> 5.1k -> inverting gain (100k || 20p) = −19.6 (-x20, Doepfer A-119) -> 1k -> OUT jack.
Consumer line (−10 dBV, 0.9 Vpp) reaches ~10 Vpp near half LEVEL; hotter sources: turn LEVEL down.

MONO: U3A sums the two gain-stage outputs (before the 1k, so a load on OUT L/R
does not move MONO): MONO = −0.51 x (GAIN_L + GAIN_R) = +10 x (L + R) at the inputs.
OUT L/R are inverted (BM topology): MONO is in phase. U3B unused: grounded follower.



Circuit intent: docs/research.md; placement: docs/geometry.json
S2 — stereo line in, OUT L/R + MONO, +/−12 V, 4HP — not released

Big Modular

Sheet: /
File: line-in.kicad_sch

Title: Big Modular — LINE IN

Size: A3	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 1/1